primary asset for United States and NATO counterterrorism operations in Afghanistan, Iraq, Syria, and Somalia.

Equipped with AGM-114 Hellfire missiles and GBU-12 Paveway II bombs, the Reaper demonstrated unmatched persistence and accuracy, enabling targeted eliminations of high-value threats while minimizing collateral damage. Its real-time data streaming capability enhanced coordination between air and ground forces, improving battlefield situational awareness. The platform's success led to widespread international adoption by countries such as the United Kingdom, Italy, and France, solidifying its reputation as a model of modern hunter-killer UAVs. However, its deployment also raised ethical and legal debates about autonomous lethal decisions and civilian impact, prompting the need for tighter operational protocols and oversight.

4.7.3 Case study 3: Dassault Neuron – European stealth UCAV demonstration (2006–2015)

The Dassault Neuron project marked Europe's first major step toward developing autonomous stealth UCAV technology. Conducted through collaboration between France, Sweden, Italy, Spain, Greece, and Switzerland, the program validated stealth design, autonomous target acquisition, and internal weapon delivery. Flight tests between 2012 and 2015 demonstrated low radar visibility, autonomous route optimization, and cooperative engagement with manned aircraft such as the Dassault Rafale. While not operationally deployed in combat, the Neuron served as a proof of concept for future European programs, including the Franco-German Future Combat Air System (FCAS). Its technological innovations, especially in fly-by-light control systems, stealth materials, and AI-driven flight management, laid the groundwork for the development of fourth-generation stealth drones with higher autonomy and mission adaptability.